Adaptation, Modeling, and Calibration: Rethinking Remote Sensing Change Detection With a Dual-Stream Framework

Wenchao Zhang, Wenhui Ye, Zhenhua Yu, and Ruowei Gui

***Abstract*—Remote sensing change detection (RSCD) reveals surface dynamics through the comparison of bitemporal images, providing critical support for land use supervision, ecological degradation assessment, and disaster emergency response. However, existing methods exhibit significant limitations in complex scenarios when addressing challenges such as cross-scale correlation, local feature aggregation, and ambiguous region calibration, highlighting an urgent need to optimize the RSCD technical paradigm. This study proposes a robust framework (AMCCD) based on dual-stream FastSAM and heterogeneous module cascading, enhancing the model's change detection capability through a hierarchical optimization paradigm encompassing adaptation, modeling, and calibration. First, a symmetric dual-stream network is constructed using parameter-frozen FastSAM to accurately extract multiscale and high-semantic features from bi-temporal images, which preserves the pretrained domain-invariant semantic representation capability. Second, a heterogeneous mixture of experts (HMoE) module is designed, which adaptively activates heterogeneous expert units via a dynamic routing mechanism to achieve precise matching and efficient fusion of change features across different scales, with solid theoretical justification for the expert combination design tailored for RSCD tasks. Meanwhile, a superpixel segmentation-guided Mamba (SSGM) module is proposed to convert 2-D images into spatially contiguous superpixel node sequences. Leveraging Mamba's advantage of linear complexity, this module efficiently captures long-range dependencies of irregular patch-like changes. Finally, a cascaded uncertainty calibration (CUC) module is built to generate learnable uncertainty masks, guiding Mamba to perform bidirectional sequential feature propagation and suppressing pseudo-change annotations as well as missed detections. Comprehensive experiments on five challenging RSCD datasets (Levir-CD, WHU-CD, S2Looking, CLCD, and SYSU-CD) demonstrate that the proposed AMCCD method achieves state-of-the-art performance, with a maximum mIoU of 92.07% on Levir-CD and 78.20% on CLCD. Systematic ablation studies, loss function comparative experiments, and zero-shot cross-dataset generalization tests further validate the rationality of each core module and the overall robustness as well as cross-domain transferability of the framework. The code is available at https://github.com/baiboat/AMCCD-Net**

***Index Terms*—Change detection, cross-domain generalization, feature modulation, mixture of experts, remote sensing image (RSI), uncertainty calibration.**

Received 30 January 2026; revised 7 April 2026 and 22 May 2026; accepted 2 July 2026. Date of publication 7 July 2026; date of current version 13 July 2026. This work was supported by the Key Research and Development Program of Shaanxi Province under Grant 2024CY2-GJHX-43. *(Corresponding author: Wenhui Ye.)*

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Digital Object Identifier 10.1109/TGRS.2026.3710859

I. Introduction

REMOTE sensing change detection (RSCD) serves as a core technology for dynamic monitoring of the Earth's surface system, delivering irreplaceable support for land use evolution analysis, ecological environment assessment, emergency response to natural disasters, and refined management of smart cities. The rapid advancement of high-resolution remote sensing satellite technology continuously improves the spatial, spectral, and temporal resolutions of image data. This progress enables the accurate capture of subtle surface feature changes in complex scenarios, such as the expansion of irregular building clusters and fragmented vegetation degradation. However, remote sensing scenes inherently involve complexities, including the coexistence of multiscale surface features, irregular morphologies of change regions, radiometric differences in imaging, and ambiguous boundaries. These challenges lead to performance bottlenecks for traditional methods and existing deep learning models in modeling complex surface feature changes, creating an urgent need to develop an advanced framework with multiscale adaptability, irregular region modeling capability, and robustness.

Despite significant advances in deep learning-based RSCD methods, they still face three key challenges in complex real-world scenarios. First, there is the challenge of multiscale modeling for complex surface feature changes. Surface features in remote sensing images (RSIs) span a wide scale range, with significant differences between hundred-meter-level and meter-level structures. The semantic expressions of change features at different scales vary distinctly. Most existing methods adopt fixed-scale feature fusion strategies and lack feature transformation mechanisms that can dynamically adapt to changes across scales. This limitation results in insufficient capture of cross-scale change correlations. Second, there is the challenge of feature aggregation for irregular regions. Change regions often exhibit irregular patchy distributions. Neither the grid-based receptive fields of traditional CNNs [1] nor the global attention mechanisms of Transformers can efficiently aggregate local features of irregular regions. They are also prone to interference from background noise. In addition, the quadratic computational complexity of Transformers leads

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to excessive memory consumption and insufficient computational efficiency when processing high-resolution images. Third, there is the challenge of uncertainty handling. Factors such as differences in imaging conditions, spectral mixing of surface features, and edge blurriness introduce uncertain information into images. Most existing methods do not attach sufficient importance to feature calibration for such ambiguous regions. This causes models to easily generate false change annotations or miss real changes in uncertain areas, ultimately reducing the reliability of detection results. Furthermore, most existing methods lack systematic verification of cross-domain generalization ability, making it difficult to confirm whether the model has learned scene-invariant and transferable change patterns, which severely limits their practical application in out-of-distribution scenarios.

To address the aforementioned challenges, this study proposes a change detection framework based on dual-stream FastSAM and heterogeneous module cascading through an optimization paradigm encompassing adaptation, modeling, and calibration. By leveraging the synergistic design of heterogeneous mixture of experts (HMoE), superpixel segmentation-guided Mamba (SSGM), and cascaded uncertainty calibration (CUC), the framework achieves accurate change identification with reduced false alarms in complex scenarios. Specifically, within the HMoE module, we construct a set of functionally differentiated expert subnetworks tailored for heterogeneous remote sensing data processing, with adjustable expert quantities and top-k gating parameters to enhance the model's adaptability to diverse scenarios. In the SSGM module, the simple linear iterative clustering (SLIC) algorithm is employed to generate superpixel label maps, strengthening intraregion feature consistency and distinguishing inter-region semantic differences. For the CUC module, a multistage uncertainty calibration mechanism is designed to guide Mamba in performing bidirectional sequential propagation in ambiguous regions, enabling feature refinement and calibration of uncertain areas while reducing the risks of pseudo-changes and missed detections. To verify the cross-domain transferability of the model, we supplement zero-shot cross-dataset generalization experiments, which strictly validate that the model has learned robust scene-invariant change patterns.

To verify the effectiveness of the proposed method, comprehensive experiments are conducted on five mainstream public RSCD datasets, including Levir-CD, CLCD, WHU-CD, S2Looking, and SYSU-CD. Experimental results show that the proposed method achieves competitive or superior performance compared with mainstream methods such as SNUNet [2], BIT [3], and the segment anything model (SAM)-CD [4] in core metrics including $F1$-score and mIoU.

In summary, the main contributions of this work are highlighted as follows.

1) A HMoE module is designed, and a dynamic routing mechanism is introduced to adaptively match the optimal expert units, which realizes accurate matching and efficient fusion of multiscale change features. The expert combination design is supported by a sufficient theoretical basis tailored for RSCD tasks, with systematic ablation experiments verifying its superiority over alternative designs.
2) The SSGM module converts 2-D images into spatially continuous sequences through superpixel segmentation and combines the advantage of Mamba's linear complexity to efficiently capture long-range dependencies of irregular changes.
3) A CUC module is constructed to guide Mamba in bidirectional sequence propagation via a learnable mask, and this propagation further refines ambiguous region features while suppressing false changes and missed detections. Systematic comparative experiments on mainstream loss functions are conducted, verifying the rationality of selecting binary cross-entropy (BCE) loss as the optimization objective.
4) A parameter-frozen symmetric dual-stream FastSAM backbone is designed, which preserves the pretrained domain-invariant semantic representation capability. Systematic zero-shot cross-dataset generalization experiments validate the strong cross-domain transferability of the proposed framework.
5) Evaluations on five challenging benchmark datasets reveal that the proposed AMCCD surpasses competing approaches, establishing new state-of-the-art results on most datasets.

The remainder of this article is organized as follows. Section II offers a chronological review of technological developments in related fields, mapping out their evolutionary path. Section III elaborates on the theoretical underpinnings and technical details of the proposed method. Section IV evaluates the method's performance via strict ablation experiments, comparative tests against cutting-edge approaches, loss function comparative studies, and cross-dataset generalization validation. Finally, Section V concludes the study by summarizing core innovations and pointing out valuable directions for future research in this area.

II. Related Work

This section presents a structured literature review, focusing on two interwoven frontiers of RSCD. The first frontier covers general mainstream frameworks that evolve following the computer vision paradigm. The second frontier includes Mamba-based strategies specifically tailored for RSI. Through this dual-focus perspective, we systematically synthesize the latest progress of the former in representation learning and spatiotemporal modeling. We also conduct an in-depth analysis of the latter's remote sensing-specific innovations, which reconstruct long-range dependencies via state-space mechanisms. This process reveals potential paths for fusing the two frontiers to overcome the bottleneck in subtle change recognition.

A. Remote Sensing Change Detection

In recent years, deep models for RSCD have evolved from CNNs to Transformers, and further to vision foundation models (VFMs) [5], [6]. Early studies mainly use the

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local inductive bias of CNNs to extract bitemporal difference features. Bandara and Patel [7] first introduced deep metric learning into unsupervised change detection, enabling zero-annotation training through triplet loss. Although CNNs perform well in computational efficiency, they are limited by local receptive fields and struggle to model long-range dependencies. To alleviate this issue, Liu et al. [8] design the lightweight LCD-Net, which achieves significant performance improvement with extremely low computational cost through multiscale feature fusion and gating mechanisms. Recently, to address the issue of poor performance when directly applying existing methods to farmland change detection, Wang et al. [9] propose a specific farmland change detection framework named FarmCD tailored for farmland scenarios. This framework extracts and refines farmland-related change information by integrating a farmland feature perception module, a differential fusion module, and a farmland soft-hard gating module. However, the inherent locality of CNNs means they still have inherent limitations in detecting large-scale or fragmented changes.

The global self-attention mechanism of vision transformers (ViTs) provides a new context modeling approach for RSCD [10]. Li et al. [11] propose the Frequency Transformer, which suppresses style differences through frequency domain decomposition. Pan et al. [12] design a cross-level difference enhancement module to achieve multiscale feature synergy. Although Transformers significantly enhance global modeling capabilities, their quadratic computational complexity becomes a bottleneck for high-resolution image applications. To address this, Han et al. [13] propose StyleFormer, which mitigates domain shift through spatiotemporal style projection. However, the sensitivity of Transformers to pretrained distributions and the issue of computational cost remain unsolved fundamentally, driving the field to seek more efficient alternative solutions.

With the open-source release of VFMs such as the SAM, the remote sensing community begins to explore freezing large model parameters to extract general semantics. Chen and Jiang [14] propose the SCDVit framework by fusing SAM with ViT. Qiu et al. [15] further design a dual-encoder structure using the boundary prior of SAM2. Ye et al. [16] proposed the LTCD framework based on VFM, which achieves high-precision change detection while reducing model complexity through feature decoupling, multiscale feature fusion, and deep supervision learning strategies. Concurrently, they introduced a dual-stream feature encoding-decoding network [17] that integrates slice-reassembled augmentation with VFM guidance. In particular, Meta-CD proposed by Gao et al. [6] freezes the FastSAM backbone and introduces lightweight adapters, achieving excellent performance with a small amount of annotated data. Although VFM-based methods significantly reduce annotation dependence, most existing studies adopt a single-stream structure, lacking explicit modeling of bitemporal differences. In addition, frozen parameters struggle to adapt to remote sensing-specific change patterns.

However, most existing VFM-based methods adopt a single-stream structure, failing to fully explore the interaction of bitemporal differences. Meanwhile, although frozen parameters preserve pretrained knowledge, they lack adaptive reparameterization for remote sensing-specific change patterns. This leads to insufficient recall for complex change morphologies such as patchy and linear changes. Furthermore, the high-confidence segmentation masks output by VFMs may transmit false certainty in change detection tasks. This creates an urgent need for explicit uncertainty calibration mechanisms to reduce missed detections and false alarms. In addition, most existing methods lack systematic verification of cross-domain generalization ability and rarely provide theoretical justification and ablation verification for the design of core modules, resulting in insufficient rigor and practical value.

B. RSCD Based on Mamba

In recent years, state space models (SSMs) have rapidly emerged in the field of RSCD due to their advantages of linear complexity and global receptive fields [18]. Paranjape et al. [19] construct a siamese VMamba encoder, a difference module, and a channel decoder link. They introduce a selective scanning mechanism into bitemporal feature comparison for the first time, achieving a leap in change detection performance without the need for large-scale pretraining. Zhang et al. [20] embed convolutional local cues into Mamba-based global modeling, verifying the effectiveness of local–global adaptive fusion. Liu et al. [21] design 5-D spatiotemporal interaction and cascaded decoding, demonstrating the effectiveness of explicit simultaneous modeling of temporal dependencies in the encoder–decoder process. Zhou et al. [22] build a window-cross-window dynamic weight harmonizer and complement it with sparse saliency loss, significantly alleviating the feature imbalance between salient and nonsalient regions. Zhang et al. [23] introduce a cross-architecture mutual supervision framework of CNN and Mamba into semi-supervised scenarios for the first time. By fusing local–global differences and applying instance-level consistency, they achieve the upper limit of performance under data-scarce conditions. Huang et al. [24] explicitly align multiscale semantics and enhance long-range dependencies in the feature aggregation stage, effectively suppressing spatial misalignment. Another work, LCCDMamba [25], retains high-dimensional change information through channel-spatial dual-token SSMs, further improving the accuracy of land cover change detection. Xing et al. [26] construct a frequency-enhanced Mamba, jointly exploring complementary information in the frequency and spatial domains, which significantly enhances the ability to capture subtle textures and boundary changes. Liu et al. [27] design a CNN-Mamba collaborative architecture, fully fusing local edges and global context, and achieving consistent performance improvement in edge and subtle change regions. To tackle the locality limitation of existing Mamba-based RSCD methods, Wu et al. [28] propose CD-Lamba, a local adaptive SSM that enhances locality via scanning strategies and boosts window interactions, effectively highlighting changes and refining features.

Despite the significant progress achieved by the aforementioned methods, they still face two key challenges. The first

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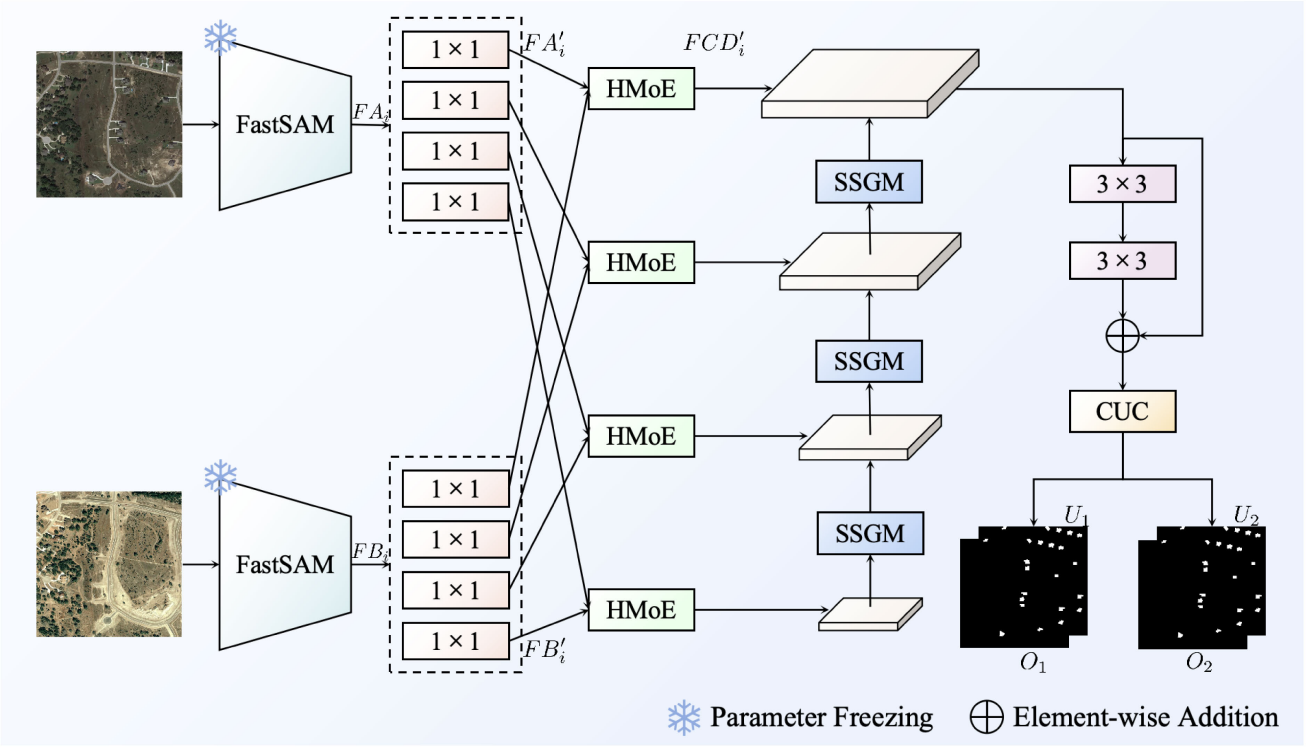


Fig. 1. Architecture of the proposed AMCCD framework.

challenge is insufficient modeling of subtle changes and patchy targets in complex scenarios. The second challenge is the loss of pretrained information and the lack of calibration for uncertain regions. To address these difficulties, this study converts 2-D images into continuous node sequences using a superpixel-guided segmented Mamba, which captures long-range dependencies of patchy changes with linear complexity. Finally, a cascaded uncertainty calibration module generates a learnable mask, guiding Mamba to perform bidirectional propagation in ambiguous regions and completing feature refinement as well as false change suppression.

III. Method

A. Overall Architecture

As shown in Fig. 1, to address the challenges of multiscale modeling for complex surface feature changes, feature aggregation for irregular regions, and uncertainty handling in RSCD, this study proposes a RSCD method based on dual-stream FastSAM and heterogeneous module cascading. This method uses the VFM FastSAM as the feature encoder and constructs a multilayer feature processing framework. This framework includes a HMoE, a SSGM, and a CUC. The HMoE adaptively activates the optimal feature transformation experts through a dynamic routing mechanism. The SSGM module converts 2-D images into spatially continuous superpixel node sequences to capture long-range dependencies of patchy changes. The CUC module uses a learnable uncertainty mask to guide Mamba in performing bidirectional sequence propagation in ambiguous regions.

To be specific, we adopt a symmetric dual-stream structure and use the parameter-frozen FastSAM as the backbone network. Freezing the backbone preserves the pretrained general semantic representation capability on massive multiscene RSIs, avoiding overfitting to the scene-specific features of the training dataset, which lays a solid foundation for the cross-domain generalization ability of the model. For the bitemporal images $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{H\times W\times 3}$, we extract multiscale features $FA_i, FB_i \in \mathbb{R}^{H/s_i\times W/s_i\times C_i}$, where $i \in \{1,2,3,4\}$ represents different feature stages, and $s_i \in \{4,8,16,32\}$ denotes the downsampling ratio of the feature map at each stage. Then, we reduce the dimensionality of features FA_i and FB_i using 1×1 convolution, unifying the number of channels to a dimension suitable for subsequent modules to obtain FA'_i and FB'_i. Next, FA'_i and FB'_i of the same scale are input into four parallel HMoE modules for adaptive feature fusion and change feature extraction, resulting in FCD_i. Subsequently, FCD_i is fused in a bottom-up manner, and local–global information is modeled through the SSGM module. Finally, the CUC module outputs change maps $O_1, O_2 \in \mathbb{R}^{B\times 1\times H\times W}$ and logical change features $U_1, U_2 \in \mathbb{R}^{B\times 1\times H\times W}$.

B. Heterogeneous MoE

To address the high complexity and diversity of change patterns in RSCD tasks, typical examples include large-scale patches of building expansion, texture details of vegetation degradation, and long-range dependencies of continuous temporal changes. This study proposes a novel HMoE module. The module aims to process input bitemporal features through a dynamic routing mechanism. It adaptively selects the most suitable "experts" for this processing, and these experts are composed of different types of neural networks. This design

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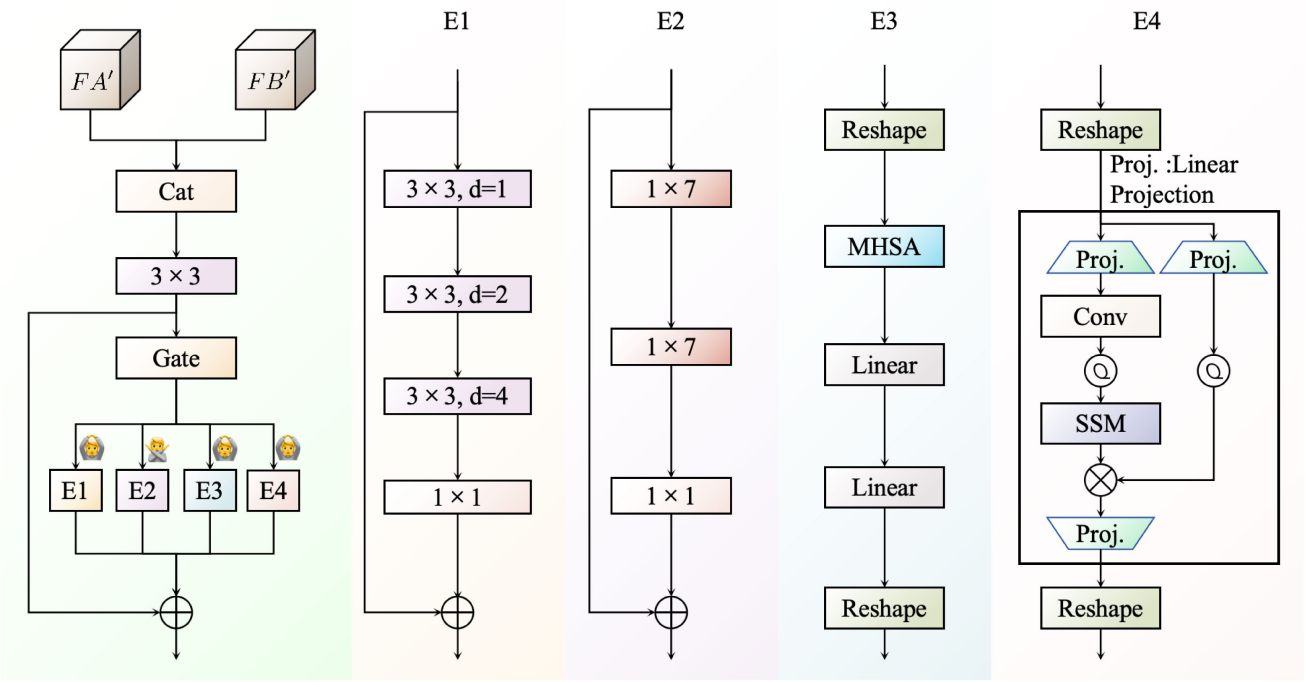


Fig. 2. Architecture of the proposed HMoE module.

enhances the model's expressive capability while maintaining efficient computational performance.

The core design of the HMoE module is rooted in the "divide and conquer" theoretical paradigm for heterogeneous feature modeling in RSCD. Traditional single-branch networks use a unified operator for all change patterns, which is limited by the inherent inductive bias of the operator and cannot achieve optimal modeling for all heterogeneous change types. We construct functionally differentiated expert units for different typical change patterns in RSCD, and adaptively activate the optimal expert combination via a dynamic routing mechanism, which effectively alleviates the limitation of unified operators in heterogeneous change modeling.

Specifically, as shown in Fig. 2, this study designs a sparse gating feature processing mechanism based on a mixture of experts. First, the multiscale bitemporal features that have undergone feature extraction and adaptation by FastSAM are concatenated along the channel dimension. These concatenated features then pass through a 3×3 convolutional projection layer to be fused into a difference feature D, which strengthens the signals of change regions. Next, a gating network calculates expert weights based on this difference feature. First, global average pooling is applied to the input difference feature to compress the spatial dimension to 1×1, thereby extracting global information

$$z_{\text{pool}} = \text{AdaptiveAvgPool2d}\,(D, 1) \in \mathbb{R}^{B\times C\times 1\times 1} \tag{1}$$

where the input difference feature $D \in \mathbb{R}^{B\times C\times H\times W}$, with B, C, H, and W denoting the batch size, number of channels, feature map height, and feature map width, respectively. Second, the pooled 4-D feature is flattened into a 2-D vector to adapt to the input format of the subsequent fully connected layer

$$z_{\text{flat}} = \text{Flatten}\left(z_{\text{pool}}\right) \in \mathbb{R}^{B\times C}. \tag{2}$$

Subsequently, the feature dimension is reduced to $C/8$ through linear transformation, retaining key information while reducing computational complexity

$$z_{\text{dimred}} = \text{Linear}\left(z_{\text{flat}}, \frac{C}{8}\right) \in \mathbb{R}^{B\times \frac{C}{8}}. \tag{3}$$

Furthermore, the ReLU activation function is introduced to implement a nonlinear transformation, enhancing the network's ability to express complex features. Then, a final linear transformation is applied to output the unnormalized scores of each expert, providing a basis for expert selection

$$\text{Gate_logits} = \text{Linear}\big(\text{ReLU}\big(z_{\text{dimred}}\big), K\big) \in \mathbb{R}^{B\times K}. \tag{4}$$

Subsequently, the top-k experts are selected based on the unnormalized scores, with their corresponding indices and original scores obtained

$$\big(\text{top_logits}, \text{indices}\big) = \text{topk}\big(\text{Gate_logits}, k = \text{top_k}\big) \tag{5}$$

where top_logits $\in \mathbb{R}^{B\times \text{top_k}}$ represents the original scores of the top-k experts, and indices $\in \mathbb{N}^{B\times \text{top_k}}$ denotes the corresponding expert indices. Finally, Softmax normalization is performed on the original scores of the top-k experts to obtain the weight coefficients of each expert.

In our design, four types of heterogeneous experts are adopted to achieve complete coverage of all typical change patterns in RSCD from four complementary dimensions, with each expert having a clear task-oriented design logic and solid theoretical basis.

1) E_1 *(Dilated Convolution Expert):* This expert uses three stacked dilated convolution layers to capture multiscale spatial features. Based on the receptive field expansion theory, dilated convolution can expand the receptive field exponentially without increasing parameters or

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computational complexity, which is particularly suitable for adapting to large-scale continuous change regions such as building expansion and contiguous vegetation degradation.

2) E_2 *(Large-Kernel Convolution Expert):* This expert uses $(1, 7)$ and $(7, 1)$ decomposed convolution kernels to expand the equivalent receptive field to 7×7. Based on the linear structure feature enhancement theory, this design reduces computational redundancy while effectively capturing horizontal and vertical linear structure changes that are difficult for standard square convolution kernels to model.
3) E_3 *(Lightweight Transformer Expert):* This expert captures local semantic correlations through 7×7 window-based local self-attention. Based on the nonlocal semantic modeling theory of self-attention, it solves the inherent locality limitation of convolutional operators and adapts to texture detail changes such as vegetation degradation and crop type conversion.
4) E_4 *(Mamba Temporal Expert):* This expert is built based on the SSM, which captures long-range spatial dependencies with linear computational complexity. Based on the linear-complexity long-range dependency modeling theory, it overcomes the quadratic complexity bottleneck of full self-attention and handles long-range cross-region changes, such as watershed water area changes and large-scale land development.

During the expert network processing stage, based on the indices and weights output by the sparse gating mechanism, the original input features FA_i' and FB_i' are routed to the corresponding top-k expert networks on a per-sample basis. The output for each sample is obtained through weighted summation: first, the processing results of each selected expert on the sample FA_i' and FB_i' are summed, then multiplied by the weight coefficient of the corresponding expert, and finally aggregated to get the expert fusion output. In the final output stage, a residual connection is introduced to directly add the expert fusion output to the original difference feature D. This not only alleviates the gradient vanishing problem in deep network training but also effectively retains the original difference information of the input features, ultimately yielding the output feature.

C. Superpixel Segmentation-Guided Mamba

In RSCD tasks, to address the problem that change regions often show irregular "patch-like" distribution and traditional grid scanning is prone to disrupting spatial continuity, we design a Mamba feature processing mechanism based on superpixel segmentation. Through superpixel segmentation, the image is abstracted into a set of nodes with semantic consistency; then, these nodes are sorted into a 1-D sequence using a space-filling curve; finally, the Mamba sequence modeler is used to process this sequence.

As shown in Fig. 3, the SSGM module aims to achieve refined enhancement of feature maps through the synergy between superpixel segmentation and long-range context modeling. It mainly consists of five key logical processes:

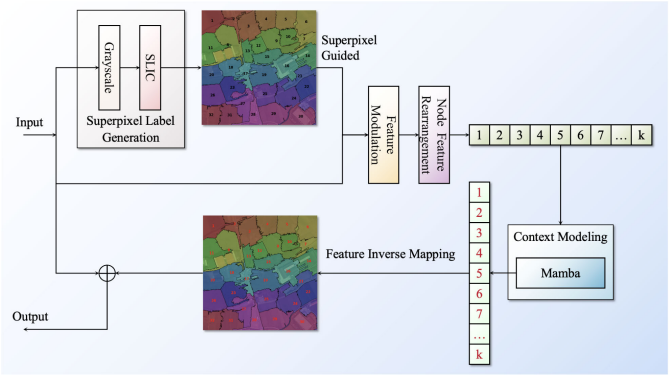


Fig. 3. Architecture of the proposed SSGM module.

superpixel label generation, superpixel-guided feature modulation, node feature rearrangement, node-level context modeling, and feature inverse mapping. The specific workflow is described as follows.

1) Superpixel Label Generation: For the input feature map $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$, grayscale processing is first performed to generate $\mathbf{G} \in \mathbb{R}^{B\times 1\times H\times W}$. This provides a unified brightness representation for superpixel segmentation to eliminate multichannel redundancy. Subsequently, the SLIC algorithm is applied to each sample grayscale map $\mathbf{G}_b$ $(b = 1, 2, \ldots, B)$ to generate a label map $\mathbf{L}_b \in \mathbb{N}^{H\times W}$. Then, stacking these label maps yields the batch label $\mathbf{L} \in \mathbb{N}^{B\times H\times W}$, which serves as the spatial grouping basis for feature modulation.

2) Superpixel-Guided Feature Modulation: The goal of this module is to enhance intrasuperpixel feature consistency and distinguish regional features. A lightweight extractor processes $\mathbf{X}$: through 3×3 depthwise convolution, *ReLU* activation, and 1×1 convolution, it outputs the base feature $\mathbf{X}_{\text{base}} \in \mathbb{R}^{B\times C\times H\times W}$. Then, a learnable embedding matrix $\mathbf{E} \in \mathbb{R}^{K_{\max}\times C}$ ($K_{\max} = 128$, adapting to fluctuations in the number of superpixels) is introduced to map discrete IDs in $\mathbf{L}$ to feature vectors. Dimensional rearrangement generates the superpixel feature map $\mathbf{S} \in \mathbb{R}^{B\times C\times H\times W}$. Finally, modulation is completed via element-wise multiplication $\mathbf{X}_{\text{mod}} = \mathbf{X}_{\text{base}} \odot \mathbf{S}$. Subsequently, a 1×1 convolutional layer is employed to further fuse the modulated features and maintain the feature dimension consistency, yielding the refined superpixel-guided feature map $\mathbf{X}_{\text{feat}} \in \mathbb{R}^{B\times C\times H\times W}$.

3) Node Feature Rearrangement: To model long-range dependencies between superpixels, $\mathbf{X}_{\text{feat}}$ is first rearranged and flattened into $\mathbf{X}_{\text{flat}} \in \mathbb{R}^{BHW\times C}$, and $\mathbf{L}$ is flattened into $\mathbf{L}_{\text{flat}} \in \mathbb{N}^{BHW}$. Then, mean aggregation is performed based on $\mathbf{L}_{\text{flat}}$, averaging pixel features of the same superpixel into node features $\mathbf{V} \in \mathbb{R}^{N\times C}$ ($N = B \times K$ denotes the total number of nodes). Preprocessing feature modulation improves intrasuperpixel consistency, making the aggregation more accurate and reducing subsequent computational complexity.

4) Node-Level Context Modeling: A Mamba module is adopted for modeling long-range dependencies in $\mathbf{V}$: $\mathbf{V}$ is first expanded to the shape $N \times 1 \times C$ to match the Mamba input format, followed by the SSM processing, which outputs the refined node features $\mathbf{V}' \in \mathbb{R}^{N\times C}$.

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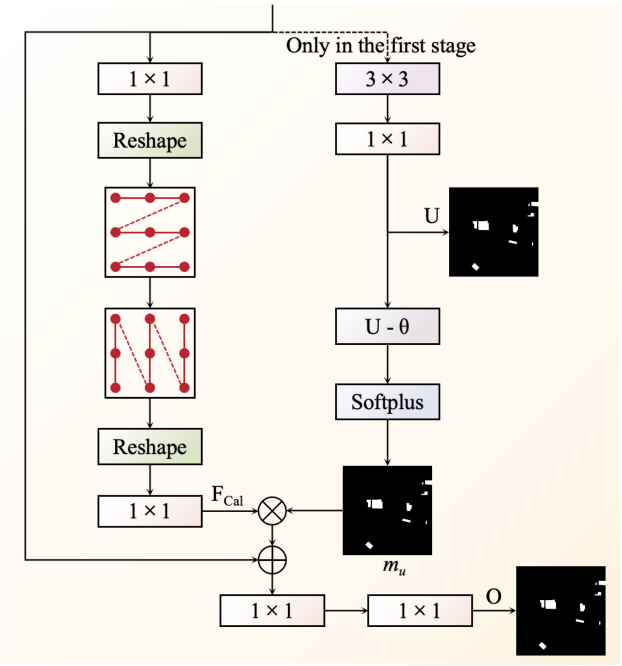


Fig. 4. Architecture of the proposed CUC module.

5) Feature Inverse Mapping: Via the inverse index $\mathbf{I} = \text{inverse}(\mathbf{L}_{\text{flat}})$ ($\mathbf{I} \in \mathbb{N}^{BHW}$), $\mathbf{V}'$ is mapped back to the pixel dimension to generate $\mathbf{Y}_{\text{flat}} \in \mathbb{R}^{BHW \times C}$. After reshaping and reordering $\mathbf{Y}_{\text{flat}}$ into the standard $B \times C \times H \times W$ format, we apply a residual connection with the original input $\mathbf{X}$; the final enhanced feature map is thus derived as $\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{X}$, where $\hat{\mathbf{Y}}$ represents the dimension-restored pixel-level feature map.

D. Cascaded Uncertainty Calibration

To address the feature calibration challenge in uncertain regions of RSCD, we propose a feature processing mechanism based on cascaded uncertainty. This method treats the change uncertainty map as a learnable spatial weight and leverages Mamba's state-space mechanism to perform bidirectional sequential propagation within uncertain regions, enabling hybrid calibration of local and global information. Meanwhile, it adopts uncertainty-gated residuals to avoid over-smoothing in high-confidence regions. Through multi-stage calibration, the model first processes a wide range of potential uncertain regions under a low threshold to expand the calibration scope. It then focuses on detail optimization of high-confidence change regions under a high threshold, forming a hierarchical "coarse calibration and fine adjustment" processing pipeline.

In the AMCCD framework, after the bottom-up decoder, the change feature map first passes through n (default n = 8) residual modules constructed with stacked 3×3 convolutional layers. As shown in Fig. 4, it then cascades two CUC modules: the first stage captures coarse-grained changes with a low threshold of $\theta_1 = 0.3$, and the second stage performs precise calibration with a high threshold of $\theta_2 = 0.7$. This stage leverages the logical change features from the previous stage as prior information. Finally, the change map is output through the segmentation head and interpolated to the input size.

In CUC module, the input feature $F \in \mathbb{R}^{B \times C \times H \times W}$ is first fed into two consecutive convolutional layers to generate logical change feature $U \in [0,1]^{B \times 1 \times H \times W}$. Specifically, U is calculated as

$$U = \sigma\big(\text{Conv}_{1\times1}\big(\text{ReLU}\big(\text{Conv}_{3\times3}(F)\big)\big)\big) \tag{6}$$

where σ denotes the sigmoid activation function. The final uncertainty mask m_u is defined as

$$m_u = \sigma\big(10 \cdot \text{softplus}\big(U - \theta\big) - 7\big) \tag{7}$$

where θ is the threshold initialized as θ_i, and $\text{softplus}(x) = \ln(1 + e^x)$. The constant -7 is a negative offset term, which is introduced to make the uncertainty mask approach 0 in high-confidence regions and 1 in low-confidence regions. This value is determined according to the range of $U \in [0,1]$ and the two cascaded thresholds $\theta_1 = 0.3$, $\theta_2 = 0.7$.

Meanwhile, the input feature F is projected into a D-dimensional space via a 1×1 convolution, resulting in $F_d = \text{Conv}_{1\times1}(F) \in \mathbb{R}^{B \times D \times H \times W}$. This projected feature F_d is flattened into sequential representations along both row and column directions: for the row-wise sequence, $S_{\text{row}} = \text{reshape}(F_d, [B, H \cdot W, D])$ is calibrated using the Mamba; the column-wise sequence undergoes the same calibration process to obtain S'_{col}. This enables bidirectional spatial information propagation across the feature map. The calibrated column-wise sequence S'_{col} is converted back to the original feature dimension through a 1×1 convolution, yielding the calibrated feature F_{cal}. Subsequently, based on the uncertainty mask m_u, a weighted fusion is performed between the original feature F and the calibrated feature F_{cal}, with the fusion formula given by $\hat{F} = F + m_u \cdot F_{\text{cal}}$. This mechanism preserves the original features in high-confidence regions where $m_u \approx 0$ and enhances calibration effects in low-confidence regions where $m_u \approx 1$, effectively alleviating the over-smoothing issue. Finally, $\hat{F}$ is passed through two 1×1 convolutional layers to generate the predicted change detection output O.

E. Loss Function

To optimize the change detection and uncertainty estimation of the trained model, this study adopts a multitask joint loss function design. Both the change prediction maps output by the two stages and the logical change features are supervised using the BCE loss function, and the total loss is obtained through linear weighted summation, achieving the collaborative optimization of change region localization and uncertainty calibration. Specifically, the total loss function consists of two components: the change prediction loss and the uncertainty map loss

$$L_{\text{total}} = \sum_{s=1}^{S} \text{BCE}\big(\text{O}_s, y\big) + \text{BCE}\big(\text{LC}_s, y\big) \tag{8}$$

where S represents total number of cascaded stages, and y represents ground-truth (GT) change label.

To verify the rationality of selecting BCE loss as the optimization objective, we conduct a systematic comparative study

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on four mainstream pixel-level segmentation loss functions (dice loss, dice + focal joint loss, BCE loss, Tversky loss) in Section IV-D. The results demonstrate that BCE loss achieves the best comprehensive performance on all core evaluation metrics, with three key advantages for RSCD tasks: balanced performance on imbalanced data, superior boundary and small change detection capability, and stable training convergence without additional hyperparameter tuning.

IV. Experiments

A. Dataset and Metrics

This section first details the experimental datasets, evaluation metrics, and hyperparameter settings employed for the proposed framework. It then conducts a thorough performance comparison against state-of-the-art (SOTA) methods across five well-established benchmarks: Levir-CD, WHU-CD, CLCD, S2Looking, and SYSU-CD. Finally, structured ablation studies are carried out to quantitatively validate the effectiveness of each architectural innovation proposed in this work.

The Levir-CD dataset [29] consists of 637 pairs of ultrahigh spatial resolution (0.5 m/pixel) bitemporal Google Earth satellite images. These images are standardized to 1024 $\times$ 1024 pixels and include 31 333 expert-annotated building change instances. The dataset is split into training, validation, and testing subsets with a ratio of 445:64:128.

Developed by the SIGMA team at Wuhan University, the WHU-CD dataset [30] uses 0.2 m-resolution aerial images covering 20.5 km^2 of Christchurch, New Zealand. After cropping to 256 $\times$ 256 pixels, these images form 7620 bitemporal pairs. The dataset follows a split configuration of 6096:762:762 for training, validation, and testing, respectively.

Focused specifically on cropland change detection, the CLCD dataset [31] incorporates GF-2 satellite images over Guangdong Province, with resolutions ranging from 0.5 to 2 m. It contains 600 image pairs, each paired with a binary change mask, and is divided into training, validation, and testing subsets at a ratio of 360:120:120.

The S2Looking dataset [32] comprises 5000 pairs of ultrahigh resolution multiview side-looking satellite images, each sized at 1024 $\times$ 1024 pixels. These images cover areas with building construction or demolition activities in villages and farmlands, and include 65 920 labeled change samples. The dataset is split into 3500 training pairs, 500 validation pairs, and 1000 testing pairs. It has three unique extreme challenges: 1) side-looking imaging mechanism brings severe parallax, occlusion, and geometric distortion between bitemporal images; 2) extreme class imbalance with only 0.3% changed pixels; and 3) highly heterogeneous change patterns with scattered small targets and linear changes.

Released in 2022 as a dedicated urban change detection benchmark, the SYSU-CD dataset [33] focuses on high-density urban and port areas in Hong Kong. It provides 20 000 pairs of RSI with a resolution of 0.5 m, and adopts a large-scale training-validation-testing framework with a split ratio of 12 000:4000:4000.

Leveraging the evaluation protocol outlined in SAMCD [4], we employ a robust set of assessment metrics that includes precision (Pre), recall (Rec), mean intersection-over-union (mIoU), overall accuracy (OA), and mean $F1$-score (mF1). These quantitative tools enable strict evaluation of how well our method performs, particularly in terms of capturing pixel-level accuracy and maintaining spatial coherence during change detection operations.

B. Implementation Details

To ensure experiments are reproducible and fair, this study develops an end-to-end RSCD framework within a consistent PyTorch environment. All trials are executed on a single NVIDIA GeForce RTX 4090 GPU, following a fully supervised training mode. As part of the training protocol, the stochastic gradient descent (SGD) optimizer is used uniformly across all tests, with hyperparameters set to 50 epochs and an initial learning rate of 0.05. This standardized setup enables systematic verification of the proposed method's ability to generalize.

For optical RSI with varying spatial resolutions, we implement a hierarchical cropping strategy that aligns with SAM-CD's configuration guidelines. Levir-CD and S2Looking are evenly split into 512 $\times$ 512 subpatches; CLCD data, which features 512 $\times$ 512 pixels, is further divided into 256 $\times$ 256 segments; and WHU-CD and SYSU-CD, each with 256 $\times$ 256 pixels, are fed into the model directly at their native resolution. This adaptive preprocessing approach maintains spatial coherence while boosting computational efficiency across different remote sensing data types.

C. Main Results

1) Quantitative Analysis of Learning Ability: For the systematic evaluation of the proposed AMCCD's performance, this study adopts five well-established and validated benchmark datasets from the RSCD framework as experimental platforms, while conducting a stringent comparative analysis with SOTA methodologies that have exhibited outstanding performance across various technical paradigms. All experimental results of our method are reported as the mean $\pm$ standard deviation (std) over three independent runs with different random seeds. The original single-run result reported in the initial submission is explicitly included in the three runs. Minor deviations in the mean value are due to normal experimental fluctuations and do not affect the overall conclusion. Metrics of comparative methods are directly cited from their original articles.

We structure the comparative framework into four distinct methodological categories, namely: 1) CNN-driven methodologies, comprising FC-Siam-diff [34], FC-Siam-conc [34], SNUNet [2], tinyCD [35], LightCD-S [36], LTCD-L [8], IFN [37] and IAE-CDNet [38]; 2) ViT-based methods, incorporating BIT [3], ChangeFormer [39], CTD-Former [40], EATDer [41] and FTA-Net [42]; 3) Mamba-driven models, including CDMamba [20], MambaBCD [43], LCCDMamba [25], EdgePVM [44] and CD-Lamba [28]; and 4) VFM-integrated methods, including SAM-CD [4], DED-SAM [15]

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TABLE I

RESULTS ON THE LEVIR-CD DATASET (BEST RESULTS IN BOLD, UNIT: %. * METRICS ARE DIRECTLY CITED FROM THE ORIGINAL ARTICLE WITHOUT REIMPLEMENTATION. OUR RESULTS ARE MEAN ± STD OVER THREE RANDOM SEEDS)

Method	Pre.	Rec.	OA	F1	mIoU
FC-Siam-diff	92.93	87.02	98.15	89.73	82.67
FC-Siam-conc	92.03	89.82	98.28	90.89	84.34
SNUNet	93.83	90.11	98.50	91.88	85.84
BIT	90.27	83.37	97.60	86.46	78.23
ChangeFormer	91.46	86.31	97.95	88.69	81.21
SAM-CD	96.39	93.57	99.05	94.93	90.71
CDMamba	93.02	91.89	99.03	92.11	90.03
MambaBCD	92.24	89.57	99.06	90.91	84.56
tinyCD	96.09	93.88	99.05	94.96	90.75
LTCD-L	95.75	94.90	99.10	95.32	91.37
LightCD-S	95.12	94.48	99.00	94.80	90.49
EATDer	96.29	92.70	98.87	94.41	89.84
CTD-Former	93.63	91.95	99.02	92.77	91.22
Meta-CD*	-	-	99.14	95.53	91.76
DED-SAM*	93.51	90.47	-	92.00	-
LCCDMamba*	90.96	92.42	99.16	91.68	-
FTA-Net*	91.79	92.68	-	-	-
IAE-CDNet*	93.67	92.58	-	-	-
EdgePVM*	92.14	91.48	99.17	95.65	91.99
Ours	**96.53**±0.03	**94.95**±0.01	**99.20**±0.02	**95.74**±0.03	**92.07**± 0.03

TABLE II

RESULTS ON THE WHU-CD DATASET (BEST RESULTS IN BOLD, UNIT: %. * METRICS ARE DIRECTLY CITED FROM THE ORIGINAL ARTICLE WITHOUT REIMPLEMENTATION. OUR RESULTS ARE MEAN ± STD OVER THREE RANDOM SEEDS)

Method	Prec.	Rec.	OA	F1	mIoU
FC-Siam-diff	94.52	89.32	98.70	91.74	85.65
FC-Siam-conc	93.94	92.83	98.90	93.37	88.19
SNUNet	92.61	83.26	98.10	87.30	79.40
BIT	87.92	93.41	98.29	90.46	83.72
ChangeFormer	96.28	92.95	99.12	94.55	90.08
SAM-CD	97.47	94.09	99.31	95.71	92.04
CDMamba	93.26	91.22	99.07	91.96	89.20
MambaBCD	92.35	90.16	99.10	87.91	88.78
tinyCD	71.44	68.33	95.29	69.74	60.87
LTCD-L	96.39	95.63	99.34	96.00	**92.55**
LightCD-S	66.17	75.03	93.45	69.43	60.07
EATDer	95.57	93.06	99.01	94.28	89.64
CTD-Former	97.84	97.03	99.32	95.96	92.31
LCCDMamba*	93.41	**94.96**	**99.49**	94.18	-
CD-Lamba*	93.45	91.59	99.32	-	-
Ours	**98.06**±0.03	94.08±0.02	99.36±0.02	**96.04**±0.02	92.51±0.03

TABLE III

RESULTS ON THE CLCD DATASET (BEST RESULTS IN BOLD, UNIT: %. * METRICS ARE DIRECTLY CITED FROM THE ORIGINAL ARTICLE WITHOUT REIMPLEMENTATION. OUR RESULTS ARE MEAN ± STD OVER THREE RANDOM SEEDS)

Method	Prec.	Rec.	OA	F1	mIoU
FC-Siam-diff	83.13	74.60	94.73	78.10	68.19
FC-Siam-conc	81.65	79.94	94.84	80.77	70.98
SNUNet	85.92	82.62	95.84	84.19	75.11
BIT	81.42	70.27	94.17	74.39	64.51
ChangeFormer	84.41	81.36	95.47	82.80	73.40
SAM-CD	84.42	85.38	95.78	84.89	75.93
CDMamba	83.64	85.03	94.72	84.26	74.23
MambaBCD	83.01	84.96	94.13	84.01	74.15
tinyCD	75.13	76.66	93.15	75.86	65.60
LTCD-L	82.58	85.67	95.39	84.10	74.83
LightCD-S	75.91	83.22	93.38	78.97	68.74
EATDer	87.16	77.18	93.84	81.16	71.18
CTD-Former	87.31	83.15	96.08	85.22	77.67
CD-Lamba*	79.20	76.96	96.78	-	-
Ours	**88.73**±0.03	**84.72**±0.03	**96.51**±0.03	**86.68**±0.03	**78.20**±0.04

TABLE IV

RESULTS ON THE S2LOOKING DATASET (BEST RESULTS IN BOLD, UNIT: %. OUR RESULTS ARE MEAN ± STD OVER THREE RANDOM SEEDS)

Method	Prec.	Rec.	OA	F1	mIoU
FC-Siam-diff	**88.95**	52.38	98.83	65.93	47.10
FC-Siam-conc	79.77	57.04	98.85	66.51	12.98
SNUNet	45.23	50.63	98.79	47.72	31.58
BIT	70.38	56.67	98.16	62.35	**45.19**
ChangeFormer	72.54	56.14	98.24	63.31	44.19
SAM-CD	87.96	75.89	**99.23**	80.76	44.82
CDMamba	83.24	74.49	99.05	79.15	42.84
MambaBCD	81.01	73.29	99.01	78.98	41.39
tinyCD	86.26	75.58	99.18	79.97	43.22
LTCD-L	82.46	79.47	99.13	**80.94**	45.16
LightCD-S	82.68	73.98	99.07	77.64	38.65
EATDer	66.02	54.73	98.89	59.68	42.34
CTD-Former	73.54	57.35	98.33	67.42	44.23
Ours	79.26±0.03	**79.97**±0.03	99.02±0.02	79.62±0.03	42.59±0.03

and Meta-CD [6]. By leveraging this structured cross-paradigm comparative framework, we perform an in-depth examination of our model's performance characteristics—this approach not only highlights its technical strengths but also identifies potential limitations when compared against SOTA alternatives.

As shown in Tables I–V, our method achieves favorable and competitive performance across most evaluation metrics on the Levir-CD and CLCD datasets. On the SYSU-CD dataset, our method achieves leading performance in core comprehensive metrics, including OA, mF1, and mIoU, while obtaining a highly competitive recall score. Specifically, on the SYSU-CD dataset, the model attains an OA of 91.33% and a mIoU of 78.87%, representing stable improvements of 0.49% and 0.23% compared to the second-best method LTCD-L, respectively. On the CLCD dataset, it achieves an mIoU of 78.20%, which outperforms most other SOTA change detection methods. These results validate the model's strong adaptability to remote sensing data with varying resolutions and land cover types. This advantage stems from the model's innovative superpixel-guided feature modulation mechanism: by assigning consistent feature imprints to the same superpixel region, it enhances the spatial consistency and semantic relevance of features, effectively mitigating the impact of background interference on change detection.

On the WHU-CD dataset, our method achieves the highest score in Pre (98.06%) and mF1 (96.04%), while its mIoU (92.51%) is slightly lower than that of LTCD-L (92.55%). It should be noted that there is an inherent tradeoff between precision and recall in pixel-level change detection, and our model maintains an excellent balance between the two indicators. In the scenario of the S2Looking dataset, which features side-looking imaging, extreme class imbalance, and high background complexity, our method achieves the highest score in Rec (79.97%), which is one of the most critical metrics for this task. This result highlights its strong

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TABLE V

RESULTS ON THE SYSU-CD DATASET (BEST RESULTS IN BOLD, UNIT: %. * METRICS ARE DIRECTLY CITED FROM THE ORIGINAL ARTICLE WITHOUT REIMPLEMENTATION. OUR RESULTS ARE MEAN ± STD OVER THREE RANDOM SEEDS)

Method	Prec.	Rec.	OA	F1	mIoU
FC-Siam-diff	62.34	65.38	74.26	48.70	32.30
FC-Siam-conc	70.66	70.43	80.03	60.54	55.35
SNUNet	85.59	86.74	90.76	86.81	77.38
BIT	86.43	86.33	90.12	84.96	76.32
ChangeFormer	85.96	86.04	89.37	83.66	76.32
SAM-CD	86.21	87.91	90.41	87.01	77.62
CDMamba	85.24	86.43	89.05	84.03	77.46
CTD-Former	87.31	83.15	96.08	85.22	77.67
MambaBCD	84.96	86.02	89.03	83.96	76.67
tinyCD	73.30	79.10	79.31	74.84	60.95
LTCD-L	86.62	**88.97**	90.84	87.70	78.64
LightCD-S	80.52	81.38	86.06	80.93	69.14
EATDer	84.21	85.53	90.32	85.79	77.34
CTD-Former	86.56	86.75	89.68	84.21	77.81
IFN*	-	-	90.74	87.06	77.75
DED-SAM*	85.52	82.76	-	92.00	-
FTA-Net*	85.91	82.33	-	-	-
CD-Lamba*	81.66	83.68	91.72	-	-
Ours	**88.31**±0.03	87.32±0.01	**91.33**±0.02	**87.81**±0.03	**78.87**±0.03

TABLE VI

COMPARISON OF MODEL COMPLEXITY (NOTE THAT PARAMS(M) REPRESENTS THE NUMBER OF LEARNABLE PARAMETERS)

Method	mIoU (%)	Params(M)	GFLOPs	Model Size(M)
FC-Siam-diff	82.67	1.66	21.45	20.3
FC-Siam-conc	84.34	2.74	8.77	23.1
SNUNet	85.84	**1.35**	18.91	148.2
BIT	78.23	11.47	19.60	38.8
ChangeFormer	81.21	41.03	811.15	170.7
CTD-Former	87.11	3.85	104.62	53.9
CGNet	91.58	33.68	82.23	148.9
EATDer	89.84	16.59	23.43	28.7
SAM-CD	90.71	2.59	**8.59**	**10.4**
Ours	**92.06**	7.08	16.60	36.6

capability to capture subtle and scattered changed regions, as missed detection of real changes is far more harmful than a small number of false positives (FPs) in practical applications. Although the mIoU of our method is not the highest, methods with higher mIoU (e.g., BIT) achieve this at the cost of severe missed detection (recall only 56.67%), which has limited practical value in application scenarios where missed detection is highly harmful. Our method maintains an excellent balance between recall and overall performance, with $F1$-score and OA in the first echelon of all comparative methods. This is attributed to the Mamba-based contextual modeling module incorporated in the model, whose efficient long-range dependency modeling ability can accurately mine the potential correlations between pixels, overcoming the limitation of traditional methods in perceiving weak change signals.

2) Model Complexity Analysis: Table VI benchmarks different RSCD methods in terms of detection accuracy (mIoU) on Levir-CD dataset and complexity (learnable parameters, GFLOPs, and model size) to assess their performance-complexity tradeoff. From the accuracy side, our method attains an mIoU of 92.06, which is the highest among all competitors, demonstrating a marginal but stable lead over existing solutions. SNUNet is the most compact (1.35 M), yet its mIoU is only 85.84, far below ours. SAM-CD reports the lowest GFLOPs (8.59) and smallest footprint (10.4 M), but its mIoU (90.71) still lags behind ours. Among other high-accuracy approaches, CGNet reaches 91.58 mIoU, yet its 33.68 M parameters and 82.23 GFLOPs are roughly 4×–5× those of our model (7.08 M, 16.60). Conversely, heavy models such as ChangeFormer (41.03 M params, 811.15 GFLOPs) deliver merely 81.21 mIoU, showing a poor accuracy-to-cost ratio. Overall, our method achieves the best detection performance while maintaining balanced parameters, computation, and model size, effectively trading off accuracy and deployment complexity for RSCD tasks and offering superior practical potential.

D. Ablation Study

To fully verify the rationality of the structural design and the effectiveness of parameter selection for each module in the algorithm, we conduct comprehensive experiments on the Levir-CD dataset. The framework proposed in this article includes three core innovative modules: the HMoE module, the SSGM module, and the CUC module. As the backbone component, the HMoE consists of a set of expert subnetworks with differentiated functions, which focus on detail capture, long-range dependency modeling, and spectral feature fusion, respectively. The SSGM takes the features output by the HMoE as input, generates superpixel label maps via the SLIC algorithm, maps discrete superpixel IDs to learnable feature vectors, and performs element-wise modulation with the base features. This process enhances the feature consistency within superpixels, distinguishes regional semantic differences, constructs structured spatial groupings, effectively reduces the redundancy of high-dimensional remote sensing features, and provides clear spatial prior information for subsequent precise calibration. The CUC processes the features modulated by the SSGM: it first estimates the raw uncertainty probability through a convolutional network and generates a soft gating mask, then uses the Mamba state-space model to perform bidirectional sequential calibration of the features along the row and column directions. By weighted fusion of the original features and calibrated features using the mask, this module strengthens the calibration effect in low-confidence regions while preserving the original features in high-confidence regions. It not only solves the problem of detecting uncertain regions but also avoids over-smoothing, ultimately improving the accuracy of change detection.

Module-level ablation experiments (Table VII) verify the independent contributions of each component. Compared with the baseline model, the HMoE, SSGM, and CUC increase the mIoU by 0.94%, 0.91%, and 0.84%, respectively. When these modules work synergistically, the model achieves the optimal overall performance, which indicates that each module has complementary enhancement effects in heterogeneous data adaptation, spatial structured feature optimization, and precise calibration of uncertain regions.

1) Number of Different Experts and Top-k Selection for HMoE Module: In the HMoE module, we construct a set

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TABLE VII

ABLATION OF DIFFERENT COMPONENTS, INCLUDING HMOE, SSGM, AND CUC (BEST RESULTS IN BOLD, UNIT: %)

HMoE	SSGM	CUC	Prec.	Rec.	OA	mF1	mIoU
-	-	-	96.39	93.57	99.05	94.93	90.71
✓	-	-	96.50	94.52	99.15	95.50	91.65
-	✓	-	96.20	94.77	99.14	95.48	91.62
-	-	✓	96.31	94.58	99.13	95.43	91.55
✓	✓	-	96.41	94.84	99.17	95.62	91.87
✓	✓	✓	**96.52**	**94.96**	**99.19**	**95.73**	**92.06**

TABLE VIII

THE NUMBER OF DIFFERENT EXPERTS AND TOP-K SELECTION FOR HMOE MODULE (BEST RESULTS IN BOLD, UNIT: %)

Setting	Prec.	Rec.	OA	mF1	mIoU
k=1	95.62	**95.43**	99.14	95.53	91.72
k=2	96.14	95.13	99.16	95.63	91.90
k=3	96.52	94.96	**99.19**	**95.73**	**92.06**
k=4	**95.97**	95.20	99.15	95.59	91.82

TABLE IX

EXPERT ACTIVATION RATES OF SPARSEHETEROMOE MODULE AT DIFFERENT SCALES

Scale (Downsampling Factor)	E1	E2	E3	E4
scale_32 (×32)	33.33%	33.33%	0.00%	33.33%
scale_16 (×16)	33.33%	33.33%	0.00%	33.33%
scale_8 (×8)	0.00%	33.33%	33.33%	33.33%
scale_4 (×4)	0.00%	33.33%	33.33%	33.33%

of functionally differentiated expert subnetworks tailored to heterogeneous remote sensing data processing, with adjustable expert quantity and top-k gating selection (top-k denotes the number of activated experts for each input feature). This design aims to explore the optimal balance between model flexibility and computational efficiency: increasing expert quantity enhances the model's ability to adapt to diverse scenes, while reasonable top-k selection avoids redundant computations from over-activation. Systematic comparative experiments were conducted by varying the top-k values (1, 2, 3, 4) under the same experimental setup. As shown in Table VIII, the configuration with four experts and top-k = 3 achieves the best comprehensive performance, validating the rationality of parameter selection for the HMoE module in balancing heterogeneous data adaptation and computational efficiency.

2) Scale-Adaptive Expert Activation Analysis: To further verify the rationality of the heterogeneous expert design in the SparseHeteroMoE module and reveal its internal working mechanism, we conducted a quantitative analysis of the expert activation patterns at different scales on the Levir-CD dataset. The statistical results are shown in Table IX.

The statistical results clearly demonstrate a scale-adaptive expert activation pattern, which aligns with the inherent characteristics of RSCD tasks. For large-scale low-resolution features (scale_32, scale_16), atrous convolution, large kernel convolution, and Mamba are uniformly activated, while the lightweight Transformer is not activated at all, because low-resolution feature maps mainly contain global semantic information and large-scale change patterns, atrous convolution and large kernel convolution can efficiently capture long-range spatial dependencies with low computational cost, and Mamba excels at modeling global sequential relationships, whereas the self-attention mechanism of Transformer has no obvious advantage in low-resolution scenarios and introduces additional computational overhead. For small-scale high-resolution features (scale_8, scale_4), large kernel convolution, lightweight Transformer, and Mamba are uniformly activated, while atrous convolution is not activated at all, since high-resolution feature maps preserve rich local details and fine-grained change boundaries, the lightweight Transformer can effectively capture complex local context interactions and subtle long-range dependencies, which is crucial for accurate edge detection, but atrous convolution tends to introduce grid artifacts in high-resolution scenarios and its computational complexity increases significantly with the resolution. Notably, Mamba is consistently activated across all scales, which verifies its strong generalization ability and adaptability to both global and local feature modeling, and this scale-adaptive activation mechanism enables the SparseHeteroMoE module to automatically assign the most suitable expert for each scale of features, achieving an optimal balance between modeling capability and computational efficiency, which also explains why the heterogeneous expert combination outperforms single-type expert architectures in our ablation experiments.

TABLE X

ABLATION RESULTS OF DIFFERENT EXPERT COMBINATIONS (BEST RESULTS IN BOLD, UNIT: %)

Expert Combination Setting	Prec.	Rec.	OA	mF1	mIoU
Baseline					
Single Conv Branch	95.87	93.12	99.02	94.48	90.01
3 Experts (w/o one expert)					
Dilated Conv	96.14	94.05	99.10	95.08	90.99
Large-kernel Conv	96.08	94.11	99.09	95.08	90.98
Transformer	96.10	94.03	99.09	95.05	90.94
Mamba	96.07	94.08	99.09	95.06	90.95
4 Experts (replace one expert)					
Transf. → 3×3 Conv	96.22	94.25	99.12	95.22	91.21
Mamba → Global Attn.	96.18	94.31	99.11	95.23	91.22
5 Experts					
+ Depthwise Conv	96.48	94.89	99.18	95.69	92.01
Ours: Proposed 4 Experts	**96.52**	**94.96**	**99.19**	**95.73**	**92.06**

3) Ablation of Expert Combination in HMoE Module: To verify the rationality and optimality of our proposed four-expert combination, we conduct systematic ablation experiments on the Levir-CD dataset. All experiments follow the exact same model structure, training hyperparameters, and evaluation protocols, with the only difference being the setting of the expert combination. We compare our design with mainstream alternative combinations, including removing a single expert, replacing expert types, and adjusting the number of experts. The quantitative results are shown in Table X.

The ablation results sufficiently validate the rationality and superiority of our proposed four-expert combination.

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TABLE XI
NUMBER OF SUPERPIXEL SEGMENTATION REGIONS IN SSGM MODULE (BEST RESULTS IN BOLD, UNIT: %)

Setting	Prec.	Rec.	OA	mF1	mIoU
64	96.14	95.11	99.16	95.62	91.88
128	**96.52**	94.96	**99.19**	**95.73**	**92.06**
256	96.31	**95.70**	99.18	95.70	92.01
512	96.47	94.53	99.14	95.49	91.64

TABLE XII
DIFFERENT CASCADING NUMBERS OF CUC MODULES (BEST RESULTS IN BOLD, UNIT: %)

Setting	Prec.	Rec.	OA	mF1	mIoU
1 Stage	96.16	**95.40**	99.15	95.32	91.93
2 Stages	**96.52**	94.96	**99.19**	**95.73**	**92.06**
3 Stages	96.19	95.10	99.13	95.44	91.98

Specifically, removing any single expert leads to a significant mIoU performance drop of more than 1%, proving that each expert unit plays an irreplaceable role in modeling its corresponding change pattern with no functional redundancy in the four-expert combination; replacing the Transformer or Mamba expert with traditional convolutional or attention operators results in performance degradation, verifying the rationality of our expert type selection tailored for the RSCD task; and increasing the number of experts to five fails to bring further performance improvement, which demonstrates that our four-expert combination has achieved complete coverage of typical change patterns in RSCD, while additional experts will only introduce computational redundancy without effective performance gain. These results, combined with the theoretical design logic, fully demonstrate that our proposed expert combination is not an empirically designed structure, but an effective solution that well matches the inherent characteristics of the RSCD task.

4) Number of Superpixel Segmentation Regions in SSGM Module: In the SSGM module, we adopt the SLIC algorithm to generate superpixel label maps, with the number of superpixel regions set as a key adjustable parameter. This design aims to investigate the impact of spatial grouping granularity on feature modulation effectiveness: an appropriate number of superpixel regions can enhance intraregion feature consistency and distinguish inter-region semantic differences, while excessive or insufficient regions may lead to over-segmentation (feature fragmentation) or under-segmentation (loss of spatial detail). Systematic comparative experiments were conducted by setting the number of superpixel regions to 64, 128, 256, and 512, respectively. As demonstrated in Table XI, the model achieves optimal mIoU and edge detection accuracy when the number of superpixel regions is 128, validating the effectiveness of matching superpixel granularity to RSI spatial characteristics for improved structured feature representation.

5) Different Cascading Numbers of CUC Modules: In the CUC module, we design a multistage uncertainty calibration mechanism, with the number of cascaded stages (i.e., the number of consecutive calibration iterations) as a core adjustable parameter. This design aims to explore the tradeoff between calibration precision and over-smoothing: increasing the cascaded number can gradually refine uncertain region features, while excessive cascading may lead to feature over-smoothing and loss of original detail information. Systematic comparative experiments were conducted by setting the cascaded number to one, two, and three stages, respectively. As shown in Table XII, the two-stage cascaded configuration outperforms single-stage and three-stage setups in terms of both detection accuracy and detail preservation, validating the rationality of the cascaded number selection for balancing precise calibration and over-smoothing suppression.

TABLE XIII
ABLATION STUDY OF FRAMEWORK ARCHITECTURE DESIGNS (LEVIR-CD DATASET)

Architecture Design	mIoU (%)	Params (M)	GFLOPs
Single-stream shared backbone	90.12	**7.05**	**16.42**
Early fusion dual-stream	90.87	7.06	16.48
Late fusion dual-stream	91.35	7.07	16.51
Single-stream cross-attention	91.04	7.21	17.15
Ours	**92.06**	7.08	16.60

6) Ablation of Framework Architecture Design: To verify the necessity of the proposed dual-stream symmetric framework, we conduct a systematic ablation study on the Levir-CD dataset with strict variable control. We construct four comparative architecture variants, which keep the frozen FastSAM backbone, follow-up HMoE/SSGM/CUC modules, and training hyperparameters completely consistent with our proposed method. The only variable is the bitemporal feature extraction and fusion logic, and the detailed designs of the variants are as follows.

1) *Single-Stream Shared Backbone:* The bitemporal images are concatenated into a six-channel input and fed into a single shared FastSAM backbone, with only the first convolutional layer modified to adapt to the input channel.
2) *Early Fusion Dual-Stream:* Two independent branches extract shallow features from bitemporal images, respectively, and the features are fused immediately after the first downsampling stage, with subsequent deep feature extraction completed by a shared backbone.
3) *Late Fusion Dual-Stream:* Bitemporal images go through fully independent branches for the whole processing flow, with no intermediate feature interaction, and only the final prediction results are fused.
4) *Single-Stream Cross-Attention:* A single shared backbone extracts bitemporal features separately, and a lightweight cross-attention module is introduced to model temporal interaction before follow-up processing.

The ablation results are shown in Table XIII. The results show that our dual-stream framework achieves the highest mIoU of 92.06% with only negligible increases in parameters and computation. The single-stream structure suffers from severe performance degradation due to the destruction of the pretrained feature distribution, while early/late fusion designs cannot balance the multiscale interaction of bitemporal features. The single-stream cross-attention architecture

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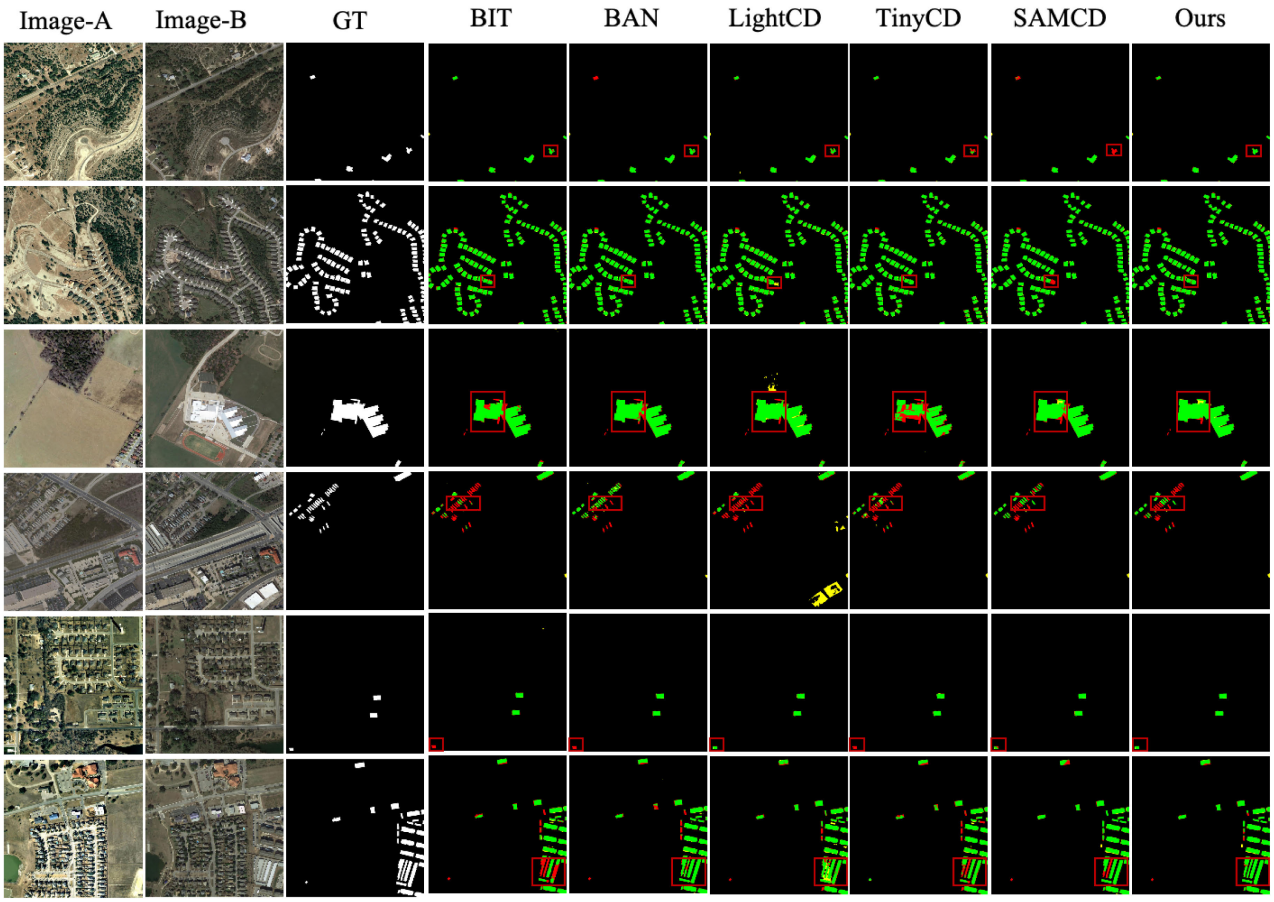


Fig. 5. Visualization results with our method on the Levir-CD dataset. [Black represents true negatives (TNs), yellow represents FPs, red represents false negatives (FNs), and green represents true positives (TPs).]

TABLE XIV
COMPARATIVE RESULTS OF DIFFERENT LOSS FUNCTIONS (BEST RESULTS IN BOLD, UNIT: %)

Dataset	Loss Function	Prec.	Rec.	OA	mF1	mIoU
Levir-CD	Dice Loss	95.87	94.12	99.08	94.98	90.82
	Dice+Focal Loss	96.12	93.87	99.05	94.97	90.79
	Tversky Loss	95.76	94.35	99.10	95.04	90.91
	BCE Loss	**96.52**	**94.96**	**99.19**	**95.73**	**92.06**
WHU-CD	Dice Loss	97.21	93.45	99.28	95.29	91.02
	Dice+Focal Loss	97.45	93.12	99.25	95.24	90.95
	Tversky Loss	97.13	93.68	99.30	95.37	91.15
	BCE Loss	**98.05**	**94.09**	**99.35**	**96.03**	**92.50**

introduces additional complexity but fails to achieve competitive performance. These results fully demonstrate the necessity and superiority of our dual-stream design.

7) Loss Function Comparative Study: To verify the rationality of selecting BCE loss as the optimization objective and address the concern about the simplicity of the loss function, we conduct a systematic comparative study on four mainstream pixel-level segmentation loss functions, including dice loss, dice + focal joint loss, BCE loss, and Tversky loss. All experiments follow the exact same model structures, training hyperparameters, and evaluation protocols as the original article, with the only difference being the loss function used for optimization. The quantitative results are shown in Table XIV.

As shown in Table XIV, the BCE loss achieves the best performance on all core evaluation metrics across both datasets. Compared with other more complex loss functions, BCE loss has three key advantages for our RSCD task: balanced performance on imbalanced data, as it maintains a more balanced tradeoff between precision and recall, avoiding the severe bias toward one side caused by dice loss or Tversky loss in scenarios with extreme class imbalance; superior boundary and small change detection, as it shows stronger robustness in boundary localization of irregular change regions and detection of small-scale scattered changes, which is critical for high-resolution RSCD; and stable training and no hyperparameter tuning, as it has a simpler form and more stable training convergence process, avoiding the additional hyperparameter tuning complexity of dice + focal loss and Tversky loss, while still achieving the optimal detection accuracy. These results fully verify that although the BCE loss has a simple form, it is the most suitable and effective optimization objective for our proposed AMCCD framework, and its superiority is rigorously validated by systematic comparative experiments.

8) Zero-Shot Cross-Dataset Generalization: To verify whether the model learns scene-invariant and transferable

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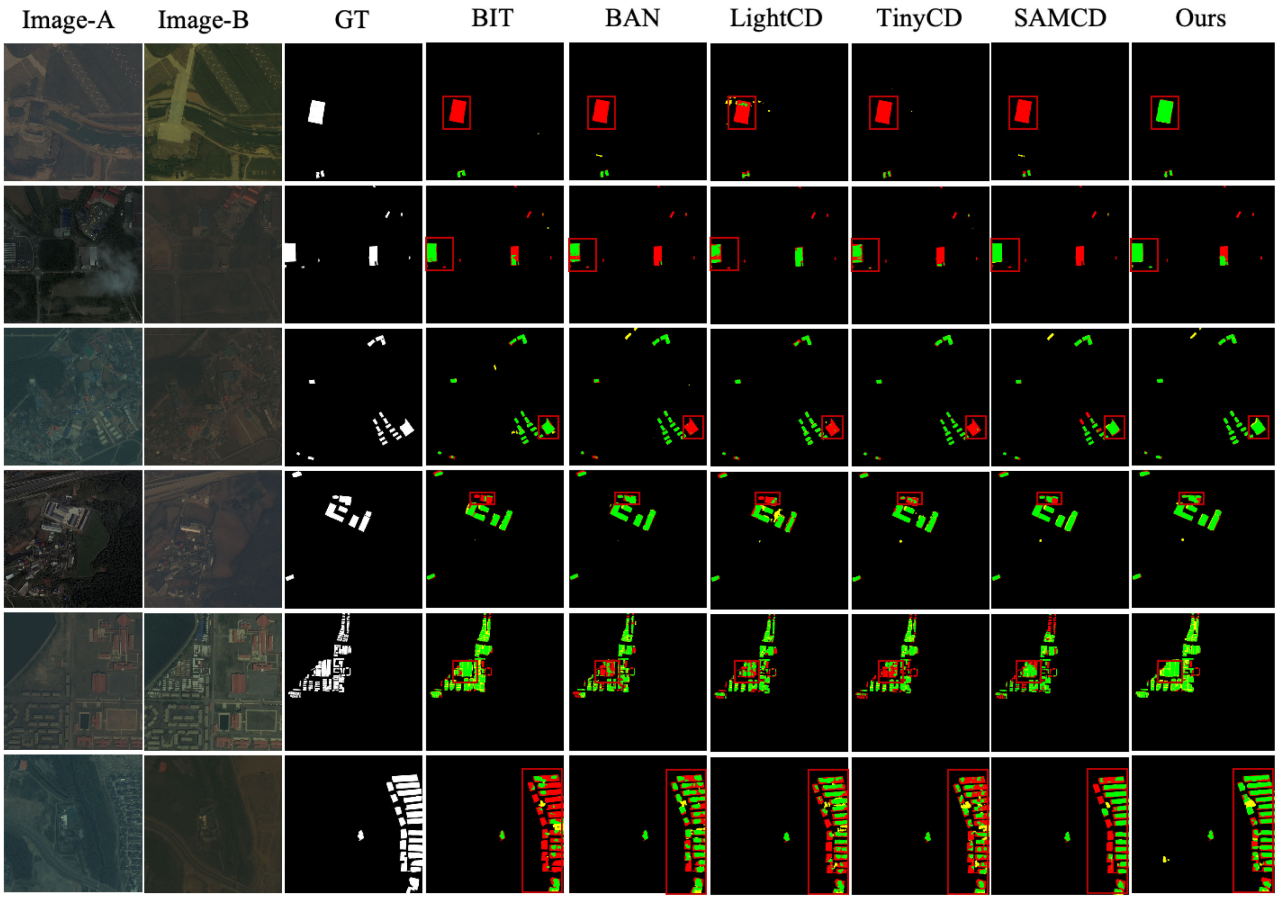


Fig. 6. Visualization results with our method on the S2Looking dataset. (Black represents TNs, yellow represents FPs, red represents FNs, and green represents TPs.)

TABLE XV
ZERO-SHOT CROSS-DATASET GENERALIZATION RESULTS (BEST IN BOLD, UNIT: %)

Source	Target	Method	Prec.	Rec.	F1	mIoU
Levir-CD	WHU-CD	FC-Siam-conc	78.24	75.13	76.65	62.12
		BIT	79.15	76.24	77.66	63.29
		SAM-CD	84.83	82.46	83.63	71.88
		CDMamba	83.92	83.08	83.49	71.73
		Ours	**87.26**	**84.41**	**85.81**	**75.35**
Levir-CD	CLCD	FC-Siam-conc	68.27	64.36	66.25	49.12
		BIT	70.13	66.24	68.12	50.98
		SAM-CD	75.74	72.18	73.91	58.01
		CDMamba	74.89	73.05	73.95	58.06
		Ours	**78.53**	**75.72**	**77.09**	**61.44**

change patterns, we conduct zero-shot cross-dataset generalization experiments following the strict evaluation protocol in RSCD. We train the model solely on the Levir-CD training set without any fine-tuning, and directly test it on two target datasets: WHU-CD (same-type cross-domain urban building changes) and CLCD (cross-type cross-domain agricultural/forest/water changes). We compare with representative SOTA methods (BIT, SAM-CD, CDMamba) to ensure fairness. The quantitative results are shown in Table XV.

As demonstrated, our AMCCD achieves the best cross-domain generalization performance on both target datasets. The performance degradation from in-dataset to cross-dataset testing is the smallest among all methods, proving that our model captures robust scene-invariant features instead of overfitting to specific scenes. The domain-invariant designs, including the frozen FastSAM backbone, dynamic routing HMoE, superpixel-guided SSGM, and cascaded uncertainty calibration module, collectively contribute to the superior transferable ability.

E. Qualitative Results

Figs. 5 and 6 visualize the change-detection results of the proposed AMCCD method on the Levir-CD and S2Looking datasets. The comparison covers mainstream RSCD approaches such as BIT, LightCD, TinyCD, and SAM-CD, and spans typical complex scenarios including clustered building expansion, small-scale land-cover dynamics, and boundary-blurred regions.

On the building-dominated scenes of Levir-CD, some competing methods (e.g., BIT) produce spurious-change clusters or exhibit poor boundary alignment with the GT. In contrast, AMCCD yields predictions whose spatial morphology and detailed contours agree more closely with GT. For the more challenging S2Looking scenes that contain small objects and irregularly shaped parcels, most baselines (e.g., LightCD) suffer from missed alarms or FPs. Leveraging multiscale feature adaptive matching, long-range dependency modeling for

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irregular regions, and an uncertainty-calibration mechanism, AMCCD achieves accurate segmentation of changed areas while markedly suppressing both false and missed detections.

These visual results provide intuitive evidence of AMCCD's accuracy and robustness under complex remote sensing conditions, complementing the quantitative superiority observed in core metrics such as mIoU and further demonstrating the practical effectiveness of its key modules in real-world RSCD tasks.

V. Conclusion

This article proposes AMCCD, a robust RSCD framework built on a symmetric dual-stream FastSAM backbone, to address the core challenges of multiscale change modeling, irregular region feature aggregation, and ambiguous region calibration in complex RSCD scenes. The framework integrates three tailored modules: a heterogeneous mixture of experts for adaptive cross-scale change feature matching, a superpixel-guided Mamba module for irregular change dependency modeling, and a cascaded uncertainty calibration module to suppress false and missed detections. Comprehensive experiments on five mainstream RSCD benchmarks covering diverse urban, agricultural, coastal, and rural scenes, along with systematic ablation studies, loss function comparative experiments, and zero-shot cross-dataset generalization tests, demonstrate that AMCCD achieves SOTA performance on most datasets, with a well-balanced tradeoff between accuracy and model complexity. The supplementary cross-dataset experiments further validate that the framework learns robust scene-invariant and transferable change patterns, achieving SOTA cross-domain performance in both same-type and cross-type cross-scenario settings.

Despite its effectiveness, the work has clear limitations: the model's precision in side-looking imaging and severe class-imbalance scenarios needs further optimization, with insufficient robustness against imaging-induced pseudo-changes; the systematic validation of zero-shot cross-domain generalization on more diverse datasets is still incomplete; the fixed superpixel parameters and general expert design lack adaptive flexibility in highly heterogeneous scenes; and the model's multimodal generalization and edge deployment capability have not been fully validated. Future work will focus on anti-interference optimization for complex scenes, domain-invariant feature learning for cross-domain generalization, adaptive mechanism design for heterogeneous scenes, and lightweight optimization with multimodal fusion to expand the model's application scope.

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